Hong Kong Physics Olympiad 2024

2024 年香港物理奧林匹克競賽

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The Hong Kong Academy for Gifted Education

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The Physical Society of Hong Kong

香港物理學會

Hong Kong Physics Olympiad Committee

香港物理奧林匹克委員會

19 May, 2024

2024 年 5 月 19 日

Rules and Regulations 競賽規則

1. All questions are in bilingual versions. You can answer in either Chinese or English, but only ONE language should be used throughout the whole paper.
所有題目均為中英對照。你可選擇以中文或英文作答，惟全卷必須以單一語言作答。

2. The multiple-choice answer sheet will be collected 1.5 hours after the start of the contest. You can start answering the open-ended questions any time after you have completed the multiple-choice questions without waiting for announcements.
選擇題的答題紙將於比賽後一小時三十分收回。若你在這之前已完成了選擇題，你亦可開始作答開放式題目，而無須等候任何宣佈。

3. On the cover of the answer book and the multiple-choice answer sheet, please write your 8-digit Contestant Number, English Name, and Seat Number.
在答題簿封面及選擇題答題紙上，請填上你的 8 位數字參賽者號碼、英文姓名、及座位號碼。

4. After you have made the choice in answering a multiple choice question, fill the corresponding circle on the multiple-choice answer sheet fully using a HB pencil.
選定選擇題的答案後，請將選擇題答題紙上相應的圓圈用 HB 鉛筆完全塗黑。

5. The open-ended problems are quite long. Please read the whole problem first before attempting to solve them. If there are parts that you cannot solve, you are allowed to treat the answer as a known answer to solve the other parts.
開放式問答題較長，請將整題閱讀完後再著手解題。若某些部分不會做，也可把它們的答案當作已知來做其他部分。

6. All numerical answers must be in (at least) 3 significant digits.
所有的數值答案必須保留至少 3 個有效數字。

The following symbols and constants are used throughout the examination paper unless otherwise specified:
除非特別注明，否則本卷將使用下列符號和常數：

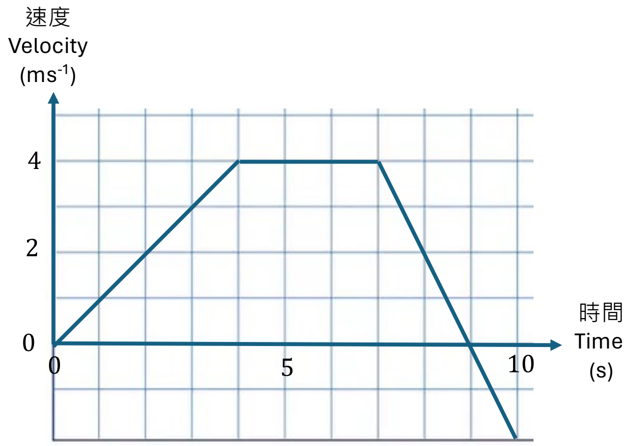
Gravitational acceleration on Earth surface 地球表面重力加速度	g	9.80 m s^{-2}
Gravitational constant 重力常數	G	$6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$

There is no friction in the problems unless otherwise specified.
除非另有說明，否則問題中沒有摩擦力。

Multiple Choice Questions (2 Marks Each) 選擇題(每題 2 分)

Below is the velocity-time graph of a toy car moving along a straight line.

以下是一輛玩具車沿著直線移動的速度-時間圖。



(1) What is the instantaneous acceleration at time $t = 8\text{s}$?

(1) 當時間 $t = 8\text{s}$ 時，瞬時加速度是多少？

A. $2\ \text{ms}^{-2}$ B. $3\ \text{ms}^{-2}$ C. $-2\ \text{ms}^{-2}$ D. $-3\ \text{ms}^{-2}$ E. $-4\ \text{ms}^{-2}$

(2) What is the displacement from the start for the toy car at $t = 10\text{s}$?

(2) 當 $t = 10s$ 時，玩具車從起點的位移是多少？

A. 20 m B. 23 m C. 25 m D. 30 m E. 32 m

Solution:

(1) acceleration at $t = 8s = slope = -\frac{6}{3} = -2ms^{-2}$

(2) The displacement is equal to the area under the curve.

$$S = \frac{1}{2}4 \times 4 + 3 \times 4 + \frac{1}{2} \times 4 \times 2 - \frac{1}{2} \times 1 \times 2 = 8 + 12 + 4 - 1 = 23$$

(3) A cannon fires projectiles at a fixed speed but with variable angle. The range of the cannon is L when it fires at an angle $30°$ above the horizontal. What is the maximum range of the cannon? You can ignore air resistance.

(3) 一門大砲以固定速率但不同角度發射砲彈。當大砲以與水平面成 $30°$ 的角度射擊時，物體的射程為 L。請問大砲的最大射程為多少？可以忽略空氣阻力。

A. L B. $\sqrt{3}L$ C. $\sqrt{2}L$ D. $2L$ E. $\frac{2}{\sqrt{3}}L$

Solution:

Assume the speed of the projectile is v,

$$0 = v\sin\frac{\pi}{6}t - \frac{1}{2}gt^2 \Rightarrow t = \frac{v}{g}$$

$$L = v\cos\frac{\pi}{6}t = \frac{\sqrt{3}}{2}vt = \frac{\sqrt{3}}{2}\frac{v^2}{g}$$

The maximum range is achieved when $\theta = \frac{\pi}{4}$,

$$v\sin\frac{\pi}{4}t-\frac{1}{2}gt^2=0\Rightarrow t=\frac{\sqrt{2}v}{g}$$

$$L_m=\frac{v}{\sqrt{2}}t=\frac{v}{\sqrt{2}}\frac{\sqrt{2}v}{g}=\frac{v^2}{g}=\frac{2L}{\sqrt{3}}$$

(4) A circular room with a diameter of 8.0 m is spinning horizontally around its center axis at a rate of 45 revolutions per minute. People are standing with their backs against the wall of the room when suddenly the floor drops out, leaving them with no support from the ground. What is the minimum coefficient of static friction between the people and the wall that is required to prevent the people from sliding down the wall?
(5) 一個直徑為 8.0 米的圓形房間以每分鐘 45 次的速度圍繞其中心軸水平旋轉。人們背靠房間的牆站立，當地面突然消失時，他們失去地面支撐。為了防止人們從牆上滑落，需要牆和人之間的最小靜摩擦係數是多少？

A. 0.0276 B. 0.0552 C. 0.110 D. 0.220 E. 2.18

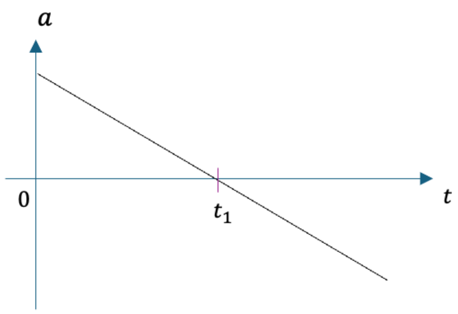
Solution:

$$N=\frac{mv^2}{R}$$

$$\mu_s N\geq f_s=mg$$

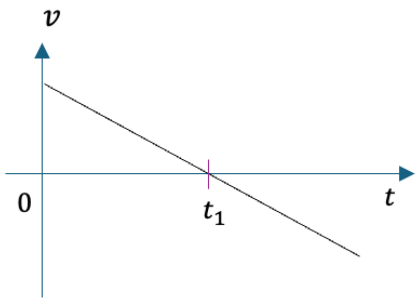
$$\Rightarrow\mu_s\geq\frac{mg}{mv^2}R=\frac{gR}{v^2}=\frac{g}{\omega^2R}=0.110$$

(5) Consider the acceleration vs. time graph of a 1D motion as
(5) 考慮一個一維運動的加速度和時間圖形如下：

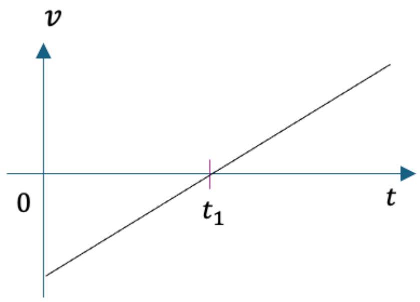


Which of the followings velocity-time graph is consistent with the acceleration-time graph above?
以下哪一個速度-時間圖與上述加速度-時間圖是一致的？

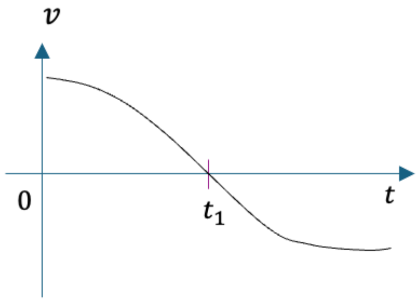
A.



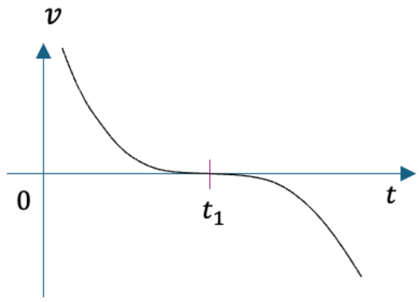
B.



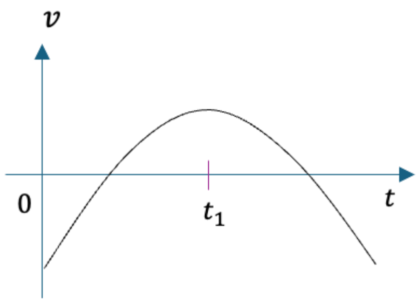
C



D.



E.



Solution:
E

The following information applies to the next two problems.
下列資訊適用於接下來的兩個問題。

An experiment involves pulling a heavy block of unknown mass across a horizontal surface with an unknown coefficient of kinetic friction. An external force is applied to the block, and the resulting acceleration of the block is shown below:
一個實驗涉及在一個水平地面上拉動一個未知質量的重物，地面與重物的動摩擦係數未知。對該重物施加一個外力，得到的加速度如下：

External Force 外力 F (N)	40	60
Acceleration 加速度 a (m/s^2)	4	8

(6) What is the value for the mass of the block?
(6) 該重物的質量是多少？

A. 1 kg B. 2 kg C. 5 kg D. 7.5 kg E. 10 kg

(7) What is the value for the coefficient of friction between the block and the surface?

(7) 該重物與表面之間的摩擦係數是多少?

A. 0.1 B. 0.2 C. 0.3 D. 0.4 E. 0.5

Solution:

$$F - \mu mg = ma$$

$$\Rightarrow F = ma + \mu mg$$

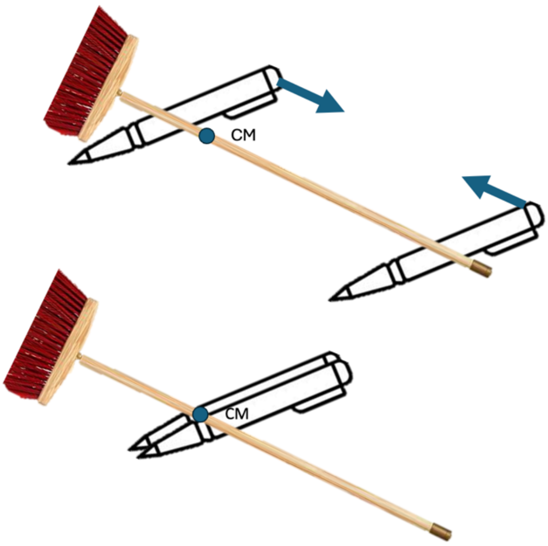
$$\Rightarrow m = slope = \frac{60-40}{8-4} = \frac{20}{4} = 5\,kg$$

$$\mu mg = intercept = F - ma = 40 - 20 = 20$$

$$\mu = \frac{20}{mg} = \frac{20}{50} = 0.4$$

(8) One can follow a few simple steps to find the center of mass of a broom. The steps are:

(8) 我們可以跟隨幾個簡單的步驟來找到掃帚的重心。這些步驟如下：



1. Hold a pen in each hand 在每隻手中拿著一支筆
2. Place the pens at each end of the broom 將筆放在掃帚的兩端
3. Lift the broom up while keeping the broom horizontal 保持掃帚水平，提起掃帚
4. Slowly move both pens towards each other until they meet 慢慢將兩支筆向彼此靠近，直到它們相遇
5. The center of mass of the broom is at the place where the pens meet. 掃帚的重心就在兩支筆相遇的地方。

To explain this trick to find the center of mass, which of the following physics is/are necessary?

為了解釋這個找重心的技巧，以下哪些物理知識是必要的？

I. It is frictionless between the broom and the pens.
掃帚和筆之間沒有摩擦。

II. The friction between the broom and the pens is linearly proportional to the normal force between them.
掃帚和筆之間的摩擦力與它們之間的法向力成線性比例。

III. Normal force between the broom and the pen is inversely proportional to the distance from the pen to the center of mass.
掃帚和筆之間的法向力與筆到重心的距離成反比。

IV. Conservation of mechanical energy.
機械能守恆。

A. I and III B. II and III C. II and IV D. I and IV E. III only

Solution: **B**

At the beginning, when one of the pen is closer to the center of mass of the broom (CM), the normal force between the broom and that pen is larger than the normal force between the broom and the other pen (further away from the CM). It is because the torques given by the normal forces with respect to the CM must be the same in magnitude:

$$n_1 r_1 = \tau = n_2 r_2$$

The normal force is inversely proportional to the distance from the pen to the CM. Since the maximum static friction is linearly proportional to the normal force between the pen and the broom, **this results in a larger maximum static friction between the broom and the pen closer to the CM.** When the pens are pushed towards each other, the pen further away from the CM will slide first because it has a smaller maximum static friction and it slides to a point until both pens are at the same distance from the CM. From that point on, in the ideal case, both pens will slides towards the CM simultaneously and meet at the CM.

(9) A boat sails downstream (same direction as the river flows) along a running river from point A. When it reaches point B, it accidentally drops a wood into the river. The boat needs to return to point A to obtain the tools to pick up the wood. It takes the same amount of time going from B to A and going from A to pick up the wood at point C in the river. Suppose the river is running at a speed $2\ \mathrm{m}/s$ relative to the ground and the boat sails from B to A and from A to C at the constant speeds, v_1 and v_2, relative to the water surface in the river, respectively. Find $v_2 - v_1$.
(9) 一艘船從上游沿著一條流動的河流（與河流流向相同）從 A 點起航。當它到達 B 點時，意外地掉了一塊木材到河裡。船需要返回 A 點取得工具來撈起木材。船從 B 點到 A 點和從 A 點到河中撈起木材的 C 點所花的時間相同。假設河流相對於地面以 2 米/秒的速度流動，船從 B 點到 A 點和從 A 點到 C 點的速度分別以相對於河水表面的定速度v_1和v_2航行。求$v_2 - v_1$。

A. 2 m/s　　B. -3 m/s　　C. 4 m/s　　D. -1 m/s　　E. 0 m/s

Solution:
E

Distance of AB = D_1

Distance of BC = D_2

Travelling time of the boat from A to B = Travelling time of the boat from A to C = T

Let the speed of the river flow is v_r

We have

From B to A: $D_1 = (v_1 - v_r)T$ --- (1)

From A to C: $D_2 = (v_2 + v_r)T$ --- (2)

The distance between B and C is $D_2 - D_1 = v_r * 2T$ --- (3)

Combining Eq. (1), (2) and (3), we have

$$v_2 - v_1 = 0$$

Alternative solution:

In the reference frame of the wood (river), the boat travels back and forth over the same distance in the same period. The boat must be travelling with the same speed with respect to the wood.

A newly discovered star system has three planets orbiting around a star at the center. The orbits of the planets are circular. Astrophysicists so far have collected the following data. Please answer questions (10) and (11) according to the table below.
一個新發現的恆星系統有三顆行星圍繞著中心的恆星軌道運行。這些行星的軌道是圓形的。到目前為止，天體物理學家收集了以下數據。請依下表回答問題 (10) 和 (11)。

Planet 行星	Planet Mass 行星質量	Orbital Radius 軌道半徑	Orbital Period 軌道週期
X	2.3×10^{23} kg	*Unknown* x 未知數 x	30 days
Y	*Unknown* y 未知數 y	6.0×10^{11} m	90 days
Z	7.1×10^{22} kg	3.5×10^{13} m	*Unknown* z 未知數 z

(10) Estimate the order of magnitude of the mass of the star at the center of the system. Assuming the star is much more massive than the planets.
(10) 估算系統中心恆星質量的數量級。假設恆星比行星質量大得多。

A. 10^{33} kg B. 10^{38} kg C. 10^{42} kg D. 10^{30} kg E. 10^{50} kg

Solution: A

Applying Kepler's 3rd Law:

$$T^2 = \frac{4\pi^2}{GM_s} r^3$$

to planet Y, we have

$$M_s = \frac{4\pi^2}{G} \frac{r^3}{T^2} = 2.11 \times 10^{33} \text{ kg}$$

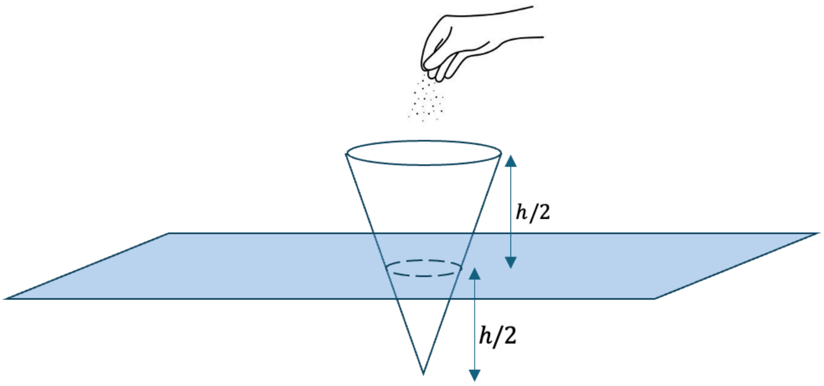
(11) Which of the following unknown(s) in the table can be determined from the existing data?
(11) 表中哪些未知數可以從現有數據找出？

A. Unknown x and y only; 僅未知數 x 和 y
B. Unknown y only; 僅未知數 y
C. Unknown x only; 僅未知數 x
D. Unknown x and z only; 僅未知數 x 和 z
E. All unknowns: x, y and z; 所有未知數：x、y 和 z

Solution: D

(12) A hollow thin cone with mass m_0 is floating stably and vertically on the water surface. When the cone is empty, half of the cone is submerged below the water level. Now sands are added into the cone. What is the mass of the sands the cone can hold before fully submerged into the water? Assume that the cone maintains vertical at any moment.

(12) 一個質量為m_0的空心薄圓錐體垂直穩定地漂浮在水面上。當圓錐體為空時，圓錐體的一半浸入水位以下。現在將沙子添加到圓錐體內。在完全浸入水中之前，圓錐體可以容納多少沙子？ 假設圓錐體一直保持垂直。



A. m_0

B. $3\,m_0$

C. $7\,m_0$

D. $9\,m_0$

E. $8\,m_0$

Solution: C

The buoyancy exerting on the cone by the water is

$$B = \rho_w g V = \rho_w g \frac{1}{3}\pi\, r^2 \frac{h}{2}$$

where r is the radius of the cone submerged under the water surface when half the cone is empty.

Initially the buoyancy cancels with the weight of the cone when it is empty; thus, we have

$$m_0 g = \rho_w g \frac{1}{3}\pi\, r^2 \frac{h}{2}$$

When the cone is just fully submerged into the water, the buoyancy is

$$B' = \rho_w g \frac{1}{3}\pi\, R^2\, h$$

According to scaling, the radius of the base of the cone and r is related by

$$R \;=\; 2r$$

The buoyancy becomes

$$B' = \rho_w g \frac{1}{3}\pi\,(2r)^2 h = 8\,B$$

To balance this buoyancy, the total mass of the cone and sands is

$$8m_0$$

Therefore, the mass of the sands is $7m_0$.

(13) A particle moves on a 2D surface under the action of a force, which always points to the right side and perpendicular to the particle's velocity. The magnitude of the force is constant. From the observer looking from above,
(13) 一顆粒子在力的作用下在 2D 表面上移動，該力始終指向右側並垂直於粒子的速度。力的大小是個常數。從觀察者的俯視角度來看，

A. the particle moves in an ellipse with constant speed. 粒子作等速率橢圓運動。
B. the particle moves in a uniform circular motion. 粒子作勻速圓週運動。
C. the particle moves in a circular with a varying speed. 粒子以隨時間改變的速率做圓週運動。
D. the particle moves in a parabola with varying speed.粒子以隨時間改變的速率沿著拋物線運動。
E. the particle could move in circular or elliptical trajectory depending on its speed. 粒子可以根據其速度沿著圓形或橢圓形軌跡移動。

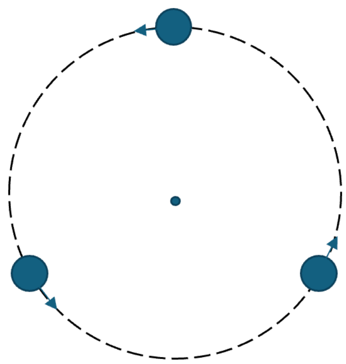
Solution: B

When a force is perpendicular to the velocity, the force only changes the direction of the motion but not the speed. The radius of curvature that the change in direction is given by

$$m\frac{v^2}{R} = F_{\perp}$$
$$\Rightarrow R = \frac{mv^2}{F_{\perp}}$$

Since the force is constant, the radius of curvature of the motion is constant; therefore, it is a uniform circular motion.

(14) Consider a triple star system in which three stars with identical mass $m = 1.00 \times 10^{30}$ kg move in the same circular orbit about their center of mass as shown in the figure. At any moment, the lines connecting the stars form an equilateral triangle. The circular orbit has a radius 1.00×10^{10} m.
(14) 考慮一個三星系統，其中三顆質量相同 $m = 1.00 \times 10^{30}$ kg 的恆星繞其質心在同一圓形軌道上運動，如圖所示。在任何時刻，連接星星的線都會形成一個等邊三角形。圓形軌道的半徑為1.00×10^{10} m。

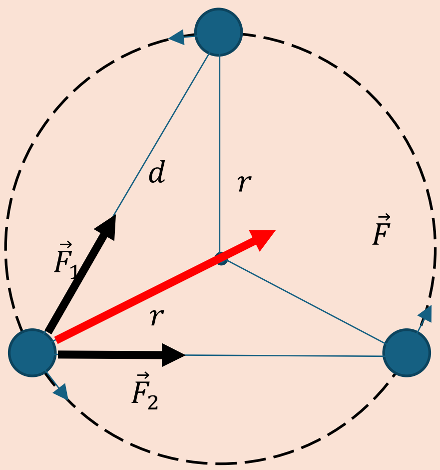


Find the period of the circular motion of one of the stars.
求其中一顆恆星的圓週運動週期。

A. 11.7 days B. 20.3 days C. 32.0 days D. 15.9 days E. 41.1 days

Solution: A

All three stars are identical. One can work out the period of one of them.



From the diagram above,

$$d^2 = r^2 + r^2 - 2r^2 \cos 120^\circ = 3r^2$$

The force acting on one of the stars is the sum of the gravitation forces due to the other two stars.

$$\vec{F} = \vec{F}_1 + \vec{F}_2$$

$$F = 2\,F_1 \cos 30^\circ = \sqrt{3}\frac{Gm^2}{d^2} = \frac{\sqrt{3}}{3}\frac{Gm^2}{r^2}$$

According to Newton's 2nd Law,

$$F = \frac{mv^2}{r} = mr\omega^2 = mr\left(\frac{4\pi^2}{T^2}\right)$$

$$\Longrightarrow \frac{\sqrt{3}}{3}\frac{Gm}{r^2} = \frac{4\pi^2 r}{T^2}$$

$$\Longrightarrow T = \sqrt{\frac{12\pi^2 r^3}{\sqrt{3}Gm}} = 1.012 \times 10^6 sec = 11.7 \text{ days}$$

(15) Consider two point particles A and B, each with the same mass m, are separated at a distance d. Particle B is initially at rest, while particle A has an initial speed v and a direction towards B. The particles are only subject to their mutual gravitational attraction. Determine the distance between the two particles at the moment when the speed of particle A is accelerated to $2v$.

(15) 考慮兩個質量相同的點粒子 A 和 B，它們相距 d。粒子 B 最初靜止，而粒子 A 具有初始速度 v，且朝向 B 的方向。這些粒子僅受到彼此之間的萬有引力作用。求粒子 A 的速度加速到 $2v$ 時，兩個粒子之間的距離。

A. $\frac{Gm}{Gm+v^2d/2}d$

B. $\frac{Gm}{Gm+3v^2d/4}d$

C. $\frac{Gm}{Gm+3v^2d/2}d$

D. $\frac{Gm}{Gm+2v^2d}d$

E. $\frac{Gm}{Gm+4v^2d}d$

Solution:

The total momentum is a conserved quantity under no external force on the whole system.

$$mv_A + mv_B = mv + 0$$

When $v_A = 2v$, we have $v_B = -v$.

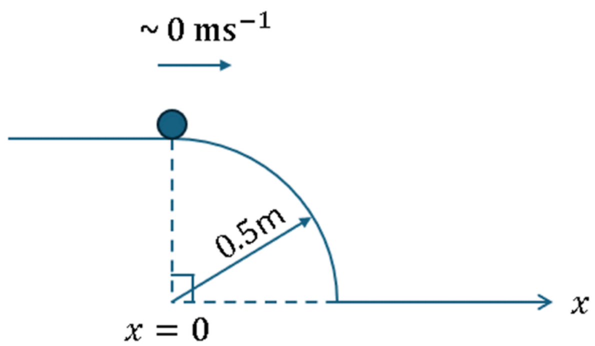
Consider conservation of energy:

$$\frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 - \frac{Gm^2}{d'} = \frac{1}{2}mv^2 + 0 - \frac{Gm^2}{d}$$

$$\frac{1}{2}4v^2 + \frac{1}{2}v^2 - \frac{Gm}{d'} = \frac{1}{2}v^2 - \frac{Gm}{d}$$

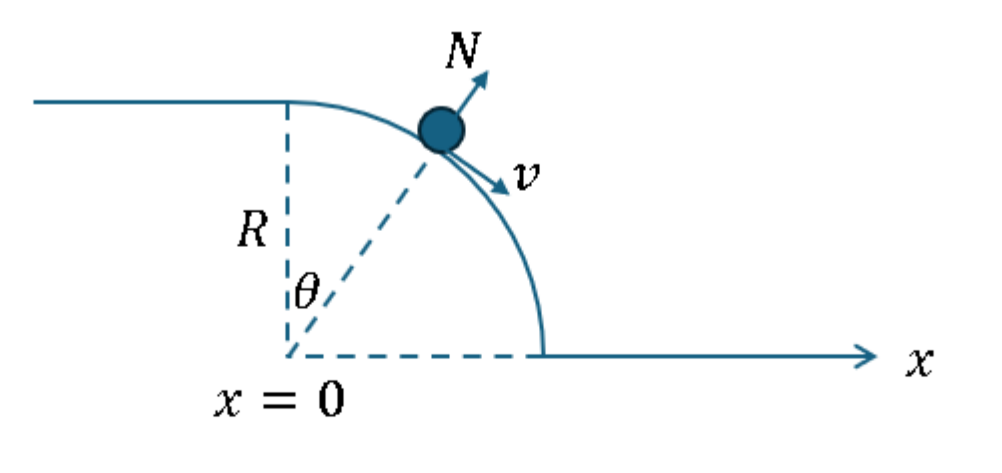
$$d' = \frac{Gm}{Gm + 2v^2 d}d$$

(16) A small ball starts to slide down a cliff, with a circular edge as shown. The ball initially travels nearly at negligible speed at $x = 0$ before sliding down the curved edge of radius 0.5 m under gravity. Determine the speed of the ball at the moment just before it leaves the curved surface.
(16) 一顆小球開始沿著懸崖滑動，懸崖有一個如圖所示的圓形邊緣。小球最初在 $x = 0$ 處以幾乎可以忽略的速度運動，然後在重力作用下沿著半徑為 0.5 m 的曲邊滑下。求小球在離開圓形邊緣前的瞬間的速度。



A. $1.28\ \mathrm{ms^{-1}}$ B. $1.69\ \mathrm{ms^{-1}}$ C. $1.81\ \mathrm{ms^{-1}}$ D. $2.21\ \mathrm{ms^{-1}}$ E. $3.13\ \mathrm{ms^{-1}}$

Solution:



Let the reaction force on the circular arc as N (in the outward radial direction).

Centripetal acceleration:

$$mg\cos\theta - N = \frac{mv^2}{R}$$

Conservation of energy

$$\frac{1}{2}mv^2 = mgR(1-\cos\theta)$$

At the point of leaving the cliff, $N = 0$, we substitute it in the above equation, and we can obtain

$$\cos\theta = \frac{2}{3} \Rightarrow \theta = \mathrm{acos}\frac{2}{3} \cong 48.19°, \qquad \sin\theta = \frac{\sqrt{5}}{3}$$

$$v = \sqrt{\frac{2gR}{3}}$$

Therefore, the ball leaves the cliff at a speed of

$$\sqrt{\frac{2\times 9.8\times 0.5}{3}} = 1.81\ ms^{-1}$$

(17) Suppose you lift up a 1-kg box on a ramp of height 1 m and constant angle, doing a work of 14.0 J, by sliding it with negligible speed on the ramp. Then, you release the box at the top of the ramp. The box slides down even though there is friction. Determine the kinetic energy of the box when it arrives the bottom of the ramp.

(17) 假設你用可忽略的速度將一個質量為 1 kg 的盒子沿著高度為 1 m 且角度恆定的斜坡抬起，做了 14.0 J 的功。然後，你在斜坡的頂端釋放盒子。儘管有摩擦，盒子將沿著斜坡滑下。求盒子到達斜坡底部時的動能。

A. 5.6 J B. 9.2 J C. 9.8 J D. 10.4 J E. 14.0 J

Solution: A

Work-done by you to lift up the block:

$$F = mg\sin\theta + f$$
$$= mg(\sin\theta + \mu\cos\theta)$$

$$W = \frac{Fh}{\sin\theta} = mgh(1 + \mu\cot\theta)$$

Therefore,

$$\mu\cot\theta = \frac{W}{mgh} - 1 = 0.429$$

Final kinetic energy W' for the block reaching the ground:

$$F' = mg(\sin\theta - \mu\cos\theta)$$
$$W' = \frac{F'h}{\sin\theta} = mgh(1 - \mu\cot\theta)$$
$$\frac{W'}{W} = \frac{1 - \mu\cot\theta}{1 + \mu\cot\theta} = 0.4$$
$$W' = 0.4 \times 14 = 5.6\,J$$

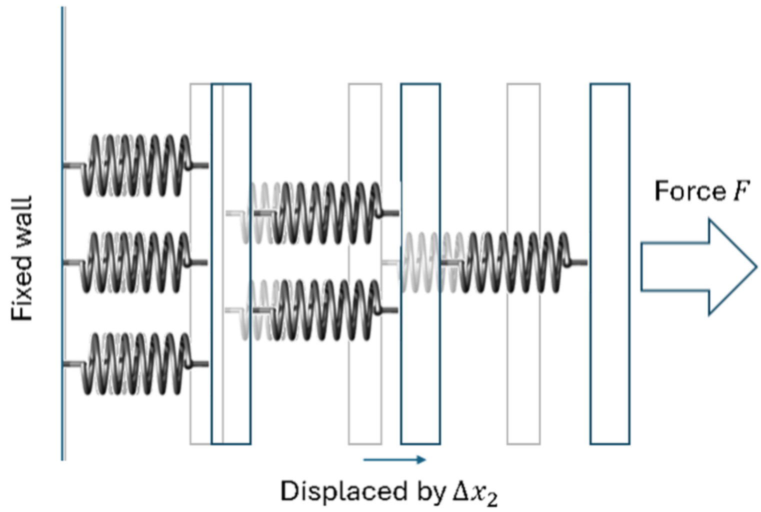
Alternative solution:

Potential energy: $U = mgh = 9.8J$

Heat: $H = 14 - 9.8 = 4.2J$

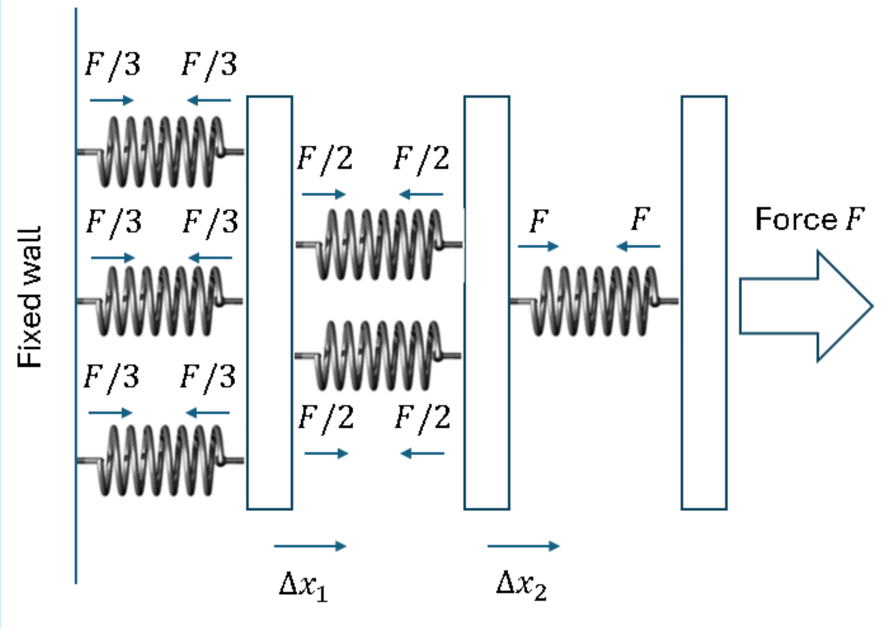
Final kinetic energy: $U - H = 9.8 - 4.2 = 5.6J$

(18) Six springs, of the same spring constant k, are assembled with 3 rigid plates as in the following figure. Now you pull the rightmost plate by a force F and the middle plate is displaced to the right by Δx_2. What is the ratio $F/\Delta x_2$?
(18) 在下圖所示的情況下，六個彈簧常數均為 k 的彈簧，與三塊剛性板組裝在一起。現在你用力 F 拉動最右邊的板，中間的板向右位移 Δx_2。 $F/\Delta x_2$ 的比值是多少？



A. $6k/11$ B. $6k/5$ C. $5k$ D. $6k$ E. None of the above

Solution: B



The force balancing diagram is listed above by assuming identical springs. Then according to Hooke's law, we have for the middle and left springs equations:

$$F/2 = k(\Delta x_2 - \Delta x_1)$$
$$F/3 = k\Delta x_1$$

Therefore

$$F/2 + F/3 = k\Delta x_2$$

and

$$F = \frac{6k}{5}\Delta x_2$$

(19) A mass hanging from the roof through a spring is undergoing simple harmonic motion of amplitude 1 mm. The spring has a spring constant of 100 N/m and the mass is 1 kg. When the displacement of the mass is at half of the amplitude, what is the speed of the mass at that moment? Please neglect the mass of the spring itself.
(19) 掛在天花板上的物塊通過彈簧進行振幅為 1 mm 的簡谐運動。彈簧的彈性常數為 100 N/m，物塊質量為 1 kg。當物塊的位移為振幅的一半時，這時物塊的速率是多少？請忽略彈簧本身的質量。

A. 5.00 mm s^{-1} B. 8.66 mm s^{-1} C. 10.00 mm s^{-1} D. 54.41 mm s^{-1} E. 62.83 mm s^{-1}

Solution:

We have the conservation of energy

$$\frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2$$

or simply

$$x^2 + v^2/\omega^2 = A^2$$

where

$$\omega = \sqrt{k/m} = 10\ s^{-1}, \qquad A = 1\ mm$$

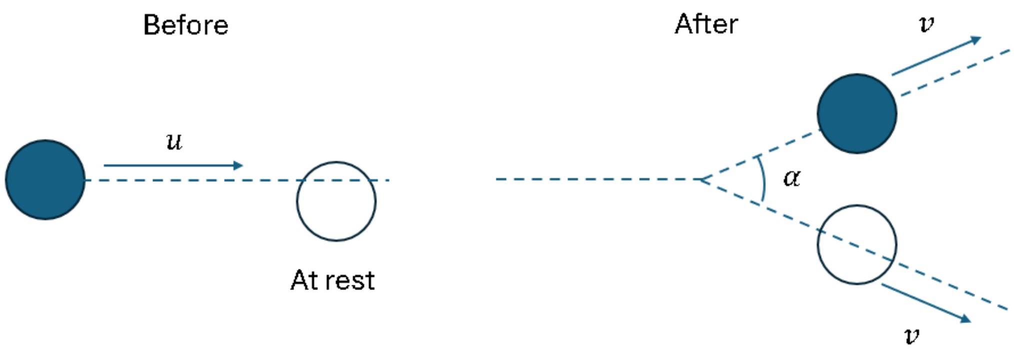
At half maximum amplitude for displacement,

$$\left(\frac{A}{2}\right)^2 + v^2/\omega^2 = A^2$$

$$v = \frac{\sqrt{3}}{2}\omega A = \frac{\sqrt{3}}{2} 10\ mms^{-1} = 8.66\ mm\ s^{-1}$$

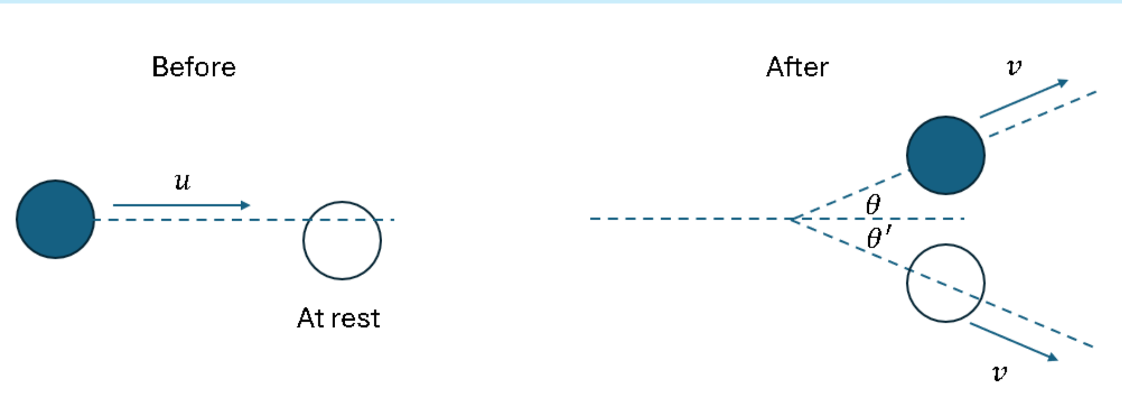
(20) Consider the elastic collision between two identical particles moving on a two-dimensional plane. The figure shows the situation before and after collisions. Before the collision, one object moving with speed u collides with another object at rest. After the collision, both objects now move at the same speed v, with an intersection angle α between the two outgoing directions. Determine α.

(20) 考慮兩個在二維平面上運動的粒子之間的彈性碰撞。圖中顯示了碰撞前後的情況。在碰撞前，一個物體以速率 u 運動，與另一個靜止的物體發生碰撞。在碰撞後，假設兩個物體現在以相同的速率 v 運動，兩個出射方向之間有一個夾角 α。確定 α 的值。



A. 60° B. 90° C. 120° D. 150° E. Cannot be determined uniquely

Solution: B



Energy conservation (given same final speed for the two objects)

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 \Rightarrow v = \frac{u}{\sqrt{2}}$$

Momentum conservation in y direction

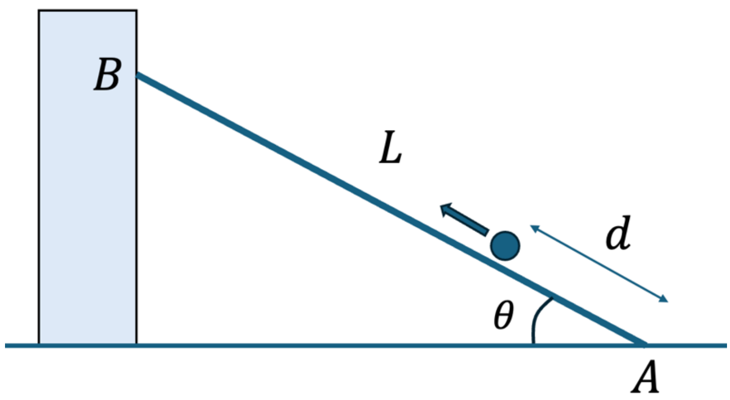
$$\theta' = \theta$$

Momentum conservation in x direction

$$mu = 2mv\cos\theta = \sqrt{2}mu\cos\theta \Rightarrow \theta = \frac{\pi}{4}$$
$$\Rightarrow \alpha = \theta + \theta' = \pi/2$$

Open Problems (20 Points Each) 開放題(每題 20 分)

1. [20] A rod with a mass of M and length L is positioned near a wall at an angle of θ with the ground. The coefficient of static friction between the rod and the ground is μ, and the friction between the rod and the wall can be neglected. A ball of mass m is moving upward along the rod with initial velocity u from point A without any friction.



1. [20] 一根質量為 M，長度為 L 的梯子以角度 θ 與地面相交，並靠近一堵牆。梯子與地面之間的靜摩擦系數為 μ，忽略梯子與牆之間的摩擦。 一顆質量為 m 的球在點 A 以初速度 u 沿著梯子向上運動，且不存在任何摩擦力。

In part (a) to (c), we are assuming that μ is sufficiently large enough so that the rod does not slide.
在(a)到(c)部分，我們假設 μ 足夠大，以至於梯子不會滑動。

(a) [2] What is the acceleration of the ball along the rod?
(a) [2] 球沿著梯子的加速度是多少？

(b) [2] What is the minimum initial velocity u such that the ball can reach the point B?
(b) [2] 如果球能夠到達點 B，初始速度 u 的最小值是多少？

(c) [3] How long does the ball take to reach the point B?
(c) [3] 球到達點 B 需要多長時間？

Now, we would like to find the minimum value of the coefficient of static friction such that the rod does not slide during the motion of the ball. We consider the moment when the ball is moving upward and located at a distance of d from point A.
現在，我們想找到最小的靜摩擦系數，使得梯子不會在球的運動過程中滑動。我們考慮球向上移動並位於距離 A 點 d的時刻。

(d) [2] What is the normal force acting on the ball?
(d) [2] 球所受的法向力是多少？

(e) [2] What is the normal force acting on the rod at point A?

(e) [2] A 點梯子所受的法向力是多少？

(f) [3] What is the normal force acting on the rod at point B?

(f) [3] B 點梯子所受的法向力是多少？

(g) [3] What is the friction force acting on the rod at point A?

(g) [3] A 點梯子所受的摩擦力是多少？

(h) [3] Find the minimum value of μ such that the rod does not slide during the motion of the ball.

(h) [3] 求 μ 的最小值是多少，以使梯子不會在球的運動期間滑動。

Solution:

(a) acceleration $= g\sin\theta$

(b) energy conservation,

$$\frac{1}{2}mu^2 > mgL\sin\theta$$

$$\Rightarrow u > \sqrt{2gl\sin\theta}$$

(c)

$$L = ut - \frac{1}{2}(g\sin\theta)t^2$$

$$\Rightarrow \frac{g\sin\theta}{2}t^2 - ut + L = 0$$

$$\Rightarrow t = \frac{u \pm \sqrt{u^2 - 2gL\sin\theta}}{g\sin\theta} = \frac{u - \sqrt{u^2 - 2gL\sin\theta}}{g\sin\theta}$$

(the positive root is neglected)

You can check that as $\theta \to 0$,

$$t = \frac{u}{g\sin\theta}\left(1 - \sqrt{1 - \frac{2gL\sin\theta}{u^2}}\right) \approx \frac{u}{g\sin\theta}\left(\frac{gL\sin\theta}{u^2}\right) = \frac{L}{u}$$

(d) The reaction force acting on the ball is

$$R = mg\cos\theta$$

(e)

$$N_A = mg\cos^2\theta + Mg$$

(f) Consider the torque with respect to point A,

$$N_B L\sin\theta = \frac{MgL}{2}\cos\theta + mg\cos\theta\, d$$

$$\Rightarrow N_B = \left(\frac{M}{2} + \frac{md}{L}\right)g\cot\theta$$

(g)

$$f + mg\cos\theta\sin\theta = N_B \Rightarrow f = N_B - mg\cos\theta\sin\theta = \left(\frac{M}{2} + \frac{md}{L} - m\sin^2\theta\right) g\cot\theta$$

(h)

$$\frac{f}{N_A} = \frac{\left(\frac{M}{2} + \frac{md}{L} - m\sin^2\theta\right)\cot\theta}{m\cos^2\theta + M} \le \mu$$

The minimum value of μ is $\frac{\left(\frac{M}{2}+m-m\sin^2\theta\right)\cot\theta}{(M+m\cos^2\theta)} = \frac{\left(\frac{M}{2}+m\cos^2\theta\right)\cot\theta}{(M+m\cos^2\theta)}$.

2. [20] A point mass m is released from rest along an incline from a height h. It slides smoothly onto a frictionless truck with a quarter-circular surface. The mass then slides along the surface and jumps vertically up with respect to the truck. Ignore all frictions in the problem.

2. [20] 一個質量 m 從高度 h 沿著斜坡從靜止釋放。它可以平穩地滑到具有四分之一圓表面的卡車上。然後，該質量沿著表面滑動並相對於卡車垂直向上躍起。忽略問題中的所有摩擦。

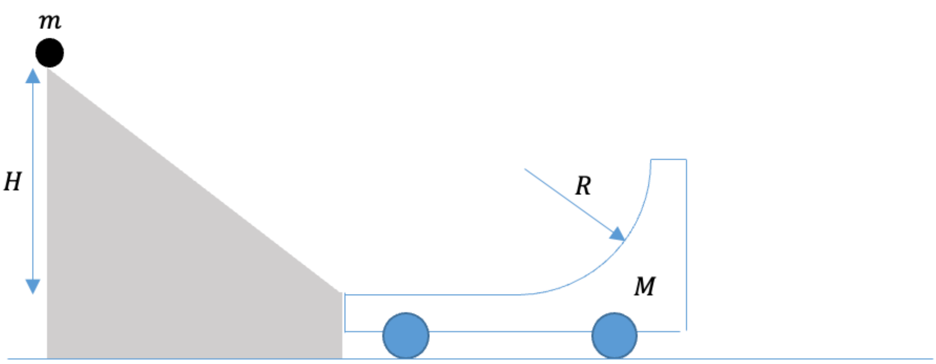


Figure 1 (a) 圖 1(a)

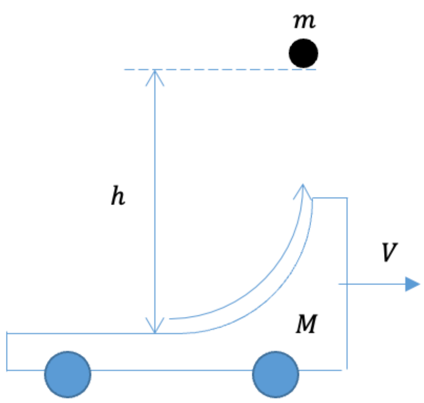


Figure 1 (b) 圖 1(b)

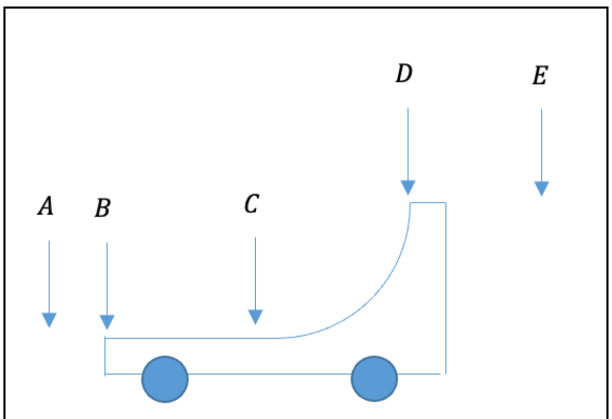


Figure 1 (c) 圖 1(c)

Given $m = 1\text{ kg},\ M = 3\text{ kg},\ R = 0.3\text{ m},\ H = 1\text{ m}$ and $g = 9.8\text{ m/s}^2$.

給與 $m = 1\text{ kg},\ M = 3\text{ kg},\ R = 0.3\text{ m},\ H = 1\text{ m}$ and $g = 9.8\text{ m/s}^2$.

(a) [6] Find the speed of the truck immediately after the mass jumps from the truck.
(a) [6]求質量從卡車上剛跳起來後卡車的速度。

(b) [6] Find the maximum height h the mass can reach, measured from the platform of the truck, after the mass jumps from the truck.
(b) [6] 求質量從卡車上跳起來後，從卡車平台開始計算，質量所能到升到的最高點高度 h。

(c) [2] After the mass reaches the highest point, it falls back down. At which part of the truck will it return to? Please write down the correct answer from the following positions, according to Figure 1 (c).
(c) [2] 當質量到達最高點後，它會落下來。它將返回卡車的哪個部分？根據圖 1 (c)，請在以下選項中選取適當的答案。

A	behind the truck 卡車後面
B	at the end of the truck 在卡車的末端
C	at the middle of the truck 在卡車的中間
D	at the front end of the quarter-circle 四分之一圓的前端
E	in front of the truck 在卡車前面

(d) [6] Find the final velocity (speed and direction) of the truck long after the mass falls back down.
(d) [6] 求質量落下來一段長時間後，卡車的最終速度（速度和方向）。

Solution:

In this problem, the total mechanical energy of the mass and the truck is conserved, while the total horizontal momentum is also conserved.

(a) When the mass is sliding down the incline, the gravitational potential energy of the mass converts into it kinetic energy. Right before the mass slides on the truck surface, it has a speed

$$\frac{1}{2}mv^2 = mgH$$
$$\Rightarrow v = \sqrt{2gH}$$

Before and after the mass slides and jumps from the truck, the total horizontal linear momentum of the mass and the truck is conserved. Therefore;

$$mv = (m + M)V$$

where V is the horizontal velocity of the truck (same as that of the mass).

$$V = \frac{m}{m+M}v = \frac{1}{4}\sqrt{2\ g\ H} = 1.107\ \mathrm{m/s}$$

(b) When the mass leaves the truck, it has a velocity with horizontal component $v_x = V = 1.107\ \mathrm{ms}^{-1}$ and a unknown vertical component v_y, but when it reaches the highest point, all of its vertical velocity is converted into the gravitational potential energy. The total mechanical energy is

$$mgH = \frac{1}{2}MV^2 + \frac{1}{2}mV^2 + mgh$$

Solving for h:

$$h = \frac{3}{4}H = 0.75 \text{ m}$$

(c) Position D.

(d) In the reference frame of the truck, none of the mass and the truck has horizontal momentum when the mass returns. The mass then slides and accelerates to the left (the end of the truck) due to normal force of the curve. At the same time the truck accelerates to the right. Right after they separate, suppose the mass and the truck have speed v' and V', respectively. According to momentum and energy conservation in this reference frame,

$$mv' = MV'$$
$$mgh = \frac{1}{2}mv'^2 + \frac{1}{2}MV'^2$$

We have

$$v' = \sqrt{\frac{2gH}{1+\frac{m}{M}}} = 3.320\frac{m}{s}$$
$$V' = 1.107\, m/s$$

Eventually, the velocity of the truck relative to the ground is $V' + V$ =1.107 + 1.107 = 2.214 m/s (to the right).

3. [20] **"Cosmic Velocities" "宇宙速度"**

In this problem, you may find the following useful:
本題可能需要用到以下數值:

Earth's Mass 地球質量	M_E	5.97×10^{24} kg
Earth's Radius 地球半徑	R	6370 km
Sun's Mass 太陽質量	M_S	1.99×10^{30} kg
Sun-Earth Average Distance (Astronomical Unit AU) 日地平均距離 (天文單位 AU)	r	1.50×10^{8} km

(a) [1] Calculate the Earth's self-rotational speed at the equator. Take the self-rotational period of the Earth to be 24 hours.
(a) [1] 計算地球在赤道處的自轉速率。 取地球自轉週期為 24 小時。

(b) [1] Assume that the Earth's orbit around the Sun is circular with radius r, calculate the orbital speed of the Earth, v_0, around the Sun.
(b) [1] 假設地球繞太陽的軌道是半徑為 r 的圓形，計算地球繞太陽的軌道速率 v_0。

In the following, we will assume that the Earth's orbit is circular. Since the Earth's self-rotational speeds are much slower than its orbital speed, they are ignored in the following calculations. It is also assumed that the Sun is at rest and the gravity of all other objects in the solar system can be ignored.
以下，我們假設地球的軌道是圓形的。 由於地球的自轉速率遠低於其軌道速率，因此在以下計算中忽略它。 同時假設太陽處於靜止狀態，並且可以忽略太陽系中所有其他物體的重力。

Define the first cosmic velocity v_1 to be the least orbital speed of a projectile launched at sea-level to keep a circular orbit around the Earth.
定義第一宇宙速度 v_1 為在海平面發射的物體保持繞地球圓形軌道的最小軌道速率。

Define the second cosmic velocity v_2, also known as the escape velocity from the Earth, to be the least speed that a projectile launched at sea-level should have so that it can escape from the Earth (which means it can reach infinite distance from the Earth when the only force acting on it is the Earth's gravitational force).).
定義第二宇宙速度 v_2，也稱為地球逃逸速度，為在海平面發射的物體能夠逃離地球（意指在只受地球引力作用下它可以到達距離地球無限遠的距離）所應具有的最小速率。

Define the third cosmic velocity v_3 to be the least speed that a projectile launched at sea-level should have so that it can escape from the solar system. The third cosmic velocity is defined <u>relative to the Earth</u> (but not the Sun). It is given that the required speed is minimum when the projectile is launched in the same direction as the orbital velocity of the Earth around the Sun.
定義第三宇宙速度 v_3 為在海平面發射的物體能夠逃離太陽系所需的最小速率。 第三宇宙速度是<u>相對於地球</u>（而不是太陽）定義的。 假設已知當物體以與地球繞太陽公轉的軌道速度相同的方向發射時，所需的速率最小。

The mass of the projectile is $m \ll M_E$.
物體的質量為 $m \ll M_E$.

It is assumed that the Earth's orbit is circular. Since the Earth's self-rotational speeds are much slower than its orbital speed, they are ignored in the following calculations. Call the projectile P and let its mass be m.

假設地球的軌道是圓形的。 由於地球的自轉速率遠低於其軌道速率，因此在以下計算中忽略它。稱發射物為P 並設其質量為 m。

It is also assumed that the Sun can be taken to be at rest because $M_S \gg M_E, m$. The gravity of all other objects in the solar system except that of the Sun, the Earth and the projectile can be ignored.
同時由於$M_S \gg M_E, m$，可以假設太陽處於靜止狀態。並且可以忽略太陽系中除太陽、地球和 P 外所有其他物體的重力。

(c) [2] Calculate the first cosmic velocity.
(c) [2] 計算第一宇宙速度。

(d) [2] Calculate the second cosmic velocity.
(d) [2] 計算第二宇宙速度。

(e) [2] Assume that the Earth's motion is not affected by its interaction with P because $M_E \gg m$. Calculate the third cosmic velocity.
(e) [2] 假設假設由於$M_E \gg m$，地球的運動不受其與 P 相互作用的影響。 計算第三宇宙速度。

The answer in (e) is actually wrong due to the wrong assumption. In the following, we will take into account the interaction between the Earth and P.
由於錯誤的假設，(e) 中的答案實際上是錯誤的。 接下來，我們將考慮地球與物體的相互作用。

(f) Let the speed of the Earth and P be v_E' and v_P', respectively, just after P is launched.
(f) 設剛發射後地球和 P 的速率分別為 v_E' 和 v_P'。

(i) [2] Obtain the relation(s) of v_0, v_E' and v_P'.
(i) [2] 寫出v_0 、v_E' 和 v_P' 的關係。

Assume the following:
During a time interval T after the launch, the Earth and P both moves approximately along a straight line perpendicular to the direction of the Sun's gravitational force. The Earth-P distance increases to many times the Earth's radius in time T so that the gravitational potential energy of the Earth-P system is negligible. The Sun-Earth and Sun-P distance both remain approximately constant in this time interval.(*)
假設下述成立：
在發射後的某時間間隔 T 內，地球和 P 均近似沿著垂直於太陽引力方向的直線運動。 地球-P 距離增加到地球半徑的許多倍，使得地球-P 系統之引力勢能可以忽略。而太陽-地球和太陽-P 間的距離在此時間間隔內近似不變。(*)

(ii) [2] Let the speed of the Earth and P be v_E'' and v_P'', respectively, at time T. Obtain the relation(s) of v_E', v_P', v_E'' and v_P''.
(ii) [2] 設 T 時刻地球和 P 的速率分別為 v_E'' 和 v_P''。 寫出 v_E' 、v_P' 、v_E'' 和 v_P'' 的關係。

(iii) [2] What is the least v_P'' so that P can escape from the solar system?
(iii) [2] 求最小 v_P'' 使得 P 能夠逃離太陽系。

(iv) [4] Hence, calculate v_3. by taking into account that $M_E \gg m$. (Note that v_3 is defined to be the speed relative to the Earth).
(iv) [4] 根據以上，並考慮$M_E \gg m$，計算 v_3。（注意 v_3 定義為相對於地球的速率）。

(g) In this part we check consistency of the answer in (f) with assumption (*).
(g) 在此部分我們檢查（f）中的答案與假設（*）是否一致。

(i) [1] Estimate T when the Earth-P distance equals 100 times the Earth's radius.
(i) [1] 估算當地球-P 距離等於地球半徑 100 倍的 T 值。

(ii) [1] Estimate the angle traversed by the Earth along its orbit around the Sun in time T.
(ii) [1] 估算地球在此時間 T 內沿著其繞太陽軌道運行過的角度。

Solution

(a)

$$\frac{2\pi}{24 \times 60 \times 60\text{s}} \times 6370\text{km} = 0.463 \text{ km/s}$$

(b)

$$v_0 = \sqrt{\frac{GM_S}{r}} \approx 29.7 \text{ km/s}$$

(c) Speed is minimum when the orbital radius is R:

$$v_1 = \sqrt{\frac{GM_E}{R}} \approx 7.91 \text{ km/s}$$

(d) By energy conservation

$$\frac{1}{2}mv_2^2 - \frac{GmM_E}{R} = 0$$

$$v_2 = \sqrt{\frac{2GM_E}{R}} = \sqrt{2}v_1 \approx 11.2 \text{ km/s}$$

(e) By energy conservation

$$\frac{1}{2}m(v_0 + v_3)^2 - \frac{GmM_E}{R} - \frac{GmM_S}{r} = 0$$

$$(v_0 + v_3)^2 - v_2^2 - 2v_0^2 = 0$$

$$v_0 + v_3 = \sqrt{v_2^2 + 2v_0^2} \approx 43.5 \text{ km/s}$$

$$v_3 = \sqrt{v_2^2 + 2v_0^2} - v_0 \approx 13.8 \text{ km/s}$$

(f) (i) By momentum conservation

$$(m + M_E)v_0 = mv_P' + M_E v_E'$$

(ii) By momentum conservation

$$mv_P' + M_E v_E' = mv_P'' + M_E v_E''$$

By energy conservation

$$\frac{1}{2}mv_P'^2 + \frac{1}{2}M_E v_E'^2 - \frac{GmM_E}{R} - \frac{GmM_S}{r} - \frac{GM_E M_S}{r} = \frac{1}{2}mv_P''^2 + \frac{1}{2}M_E v_E''^2 - \frac{GmM_S}{r} - \frac{GM_E M_S}{r}$$

$$\frac{1}{2}mv_P'^2 + \frac{1}{2}M_E v_E'^2 - \frac{GmM_E}{R} = \frac{1}{2}mv_P''^2 + \frac{1}{2}M_E v_E''^2$$

(iii) By energy conservation

$$\frac{1}{2}mv_P''^2 - \frac{GmM_S}{r} = 0$$

$$v_P'' = \sqrt{\frac{2GM_S}{r}} = \sqrt{2}v_0$$

(iv)

$$\begin{cases} (m+M_E)v_0 = mv_P' + M_E v_E' = mv_P'' + M_E v_E'' \\ \frac{1}{2}mv_P'^2 + \frac{1}{2}M_E v_E'^2 - \frac{GmM_E}{R} = \frac{1}{2}mv_P''^2 + \frac{1}{2}M_E v_E''^2 \end{cases}$$

$$\begin{cases} (m+M_E)v_0 = mv_P' + M_E v_E' = \sqrt{2}mv_0 + M_E v_E'' \\ mv_P'^2 + M_E v_E'^2 - mv_2^2 = 2mv_0^2 + M_E v_E''^2 \end{cases}$$

$$mv_P'^2 + \frac{1}{M_E}[(m+M_E)v_0 - mv_P']^2 - mv_2^2 = 2mv_0^2 + \frac{1}{M_E}\left[(m+M_E)v_0 - \sqrt{2}mv_0\right]^2$$

$$M_E v_P'^2 - 2(m+M_E)v_0 v_P' + mv_P'^2 - M_E v_2^2 = 2M_E v_0^2 - 2\sqrt{2}(m+M_E)v_0^2 + 2mv_0^2$$

$$(m+M_E)v_P'^2 - 2(m+M_E)v_0 v_P' - M_E(v_2^2 + 2v_0^2) + 2\sqrt{2}(m+M_E)v_0^2 - 2mv_0^2 = 0$$

$$v_P'^2 - 2v_0 v_P' - \frac{M_E}{m+M_E}(v_2^2 + 2v_0^2) + 2\sqrt{2}v_0^2 - \frac{2m}{m+M_E}v_0^2 = 0$$

Since $m \ll M_E$,

$$v_P'^2 - 2v_0 v_P' - (v_2^2 + 2v_0^2) + 2\sqrt{2}v_0^2 = 0$$

$$(v_P' - v_0)^2 - v_2^2 - 3v_0^2 + 2\sqrt{2}v_0^2 = 0$$

$$(v_P' - v_0)^2 = v_2^2 + \left(3 - 2\sqrt{2}\right)v_0^2 = v_2^2 + \left(\sqrt{2} - 1\right)^2 v_0^2$$

$$v_3 = v_P' - v_0 = \sqrt{v_2^2 + \left(\sqrt{2} - 1\right)^2 v_0^2} \approx 16.6 \text{ km/s}$$

(g) (i) Let's estimate the time for the distance to increase to

$100R \approx 640000$ km. The relative speed changes from 16.6 km/s to 12.3 km/s. So

The time taken is

$$\approx \frac{100R}{16.6 \text{ km/s}} \approx 10 \text{ hours}$$

(ii) The angle traversed is very small because

$$\frac{10 \text{ hours}}{1 \text{ year}} \approx 0.4°$$

The projectile traversed a slightly larger angle but same order of magnitude.

~ END of PAPER 全卷完 ~